

MODEL SHOWDOWN

Predicting company bankruptcy risk

KNN vs Logistic Regression vs Random Forest

SDG 8

Decent Work and Economic Growth

A model comparison for
early, voluntary support

1. The problem

A concrete SDG 8 issue, supported by evidence

122

enterprise bankruptcies
in Poland · Q1 2023

+38.6%

vs. Q1 2022

What we want to support

Use financial ratios to identify companies with signs associated with later bankruptcy, so an SME-support organisation can consider offering help earlier.

Why SDG 8?

Business failure can threaten jobs, household income and local economic resilience.

Evidence: Statistics Poland, 2023

2. The user and decision

A named user, a specific decision, and clear boundaries

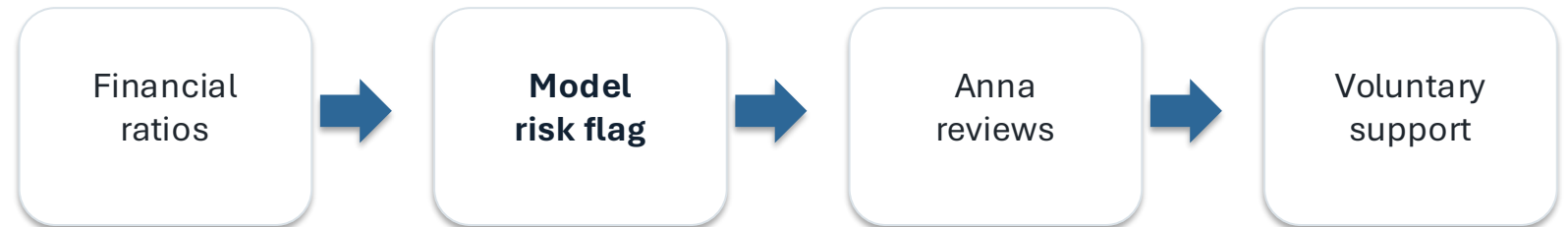
**ANNA
KOWALSKA**

Fictional financial-
resilience adviser

Polish regional
SME-support organisation

Her decision

Which companies should be considered for a
confidential, voluntary financial-health consultation?



NOT for loans, pricing, collections, procurement or public disclosure.

Not represented well enough: current non-Polish companies; demographic fairness cannot be established from this dataset.

3. Data → model → prediction

A reproducible scikit-learn pipeline

DATASET

**UCI Polish
Companies
Bankruptcy Data**

5,910 rows

64 financial ratios
+ bankruptcy label

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Historical Polish data

1 Split first 80% train / 20% test

2 Preprocess Median imputation + scaling

3 Tune 5-fold stratified CV

4 Evaluate Precision · Recall · F1

5 Use New case → flag + probability

Test set stays untouched until final evaluation.

4. Fair comparison

Same split, same folds, same main metric

MAIN METRIC

RECALL

A false negative means an at-risk company is missed.

Bankruptcy class in training data

6.97%

KNN

Best parameters

n_neighbors = 3
p = 2 · uniform

CV recall **0.163 ± 0.043**

Logistic Regression

Best parameters

C = 100
class_weight = balanced

CV recall **0.712 ± 0.031**

Random Forest

Best parameters

200 trees · depth 12
min leaf 10 · balanced

CV recall **0.518 ± 0.045**

Baseline: DummyClassifier (most frequent) → CV recall 0.000 ± 0.000

5. MODEL SHOWDOWN

The winner is chosen on our pre-selected metric: recall

Model	Best parameters	CV recall	Test precision	Test recall	Test F1
Dummy	most frequent	0.000 ± 0.000	0.000	0.000	0.000
KNN	k=3 · p=2 · uniform	0.163 ± 0.043	0.405	0.183	0.252
LOGISTIC REGRESSION	C=100 · balanced	0.712 ± 0.031	0.223	0.744	0.343
Random Forest	200 · depth 12 · leaf 10	0.518 ± 0.045	0.480	0.573	0.522

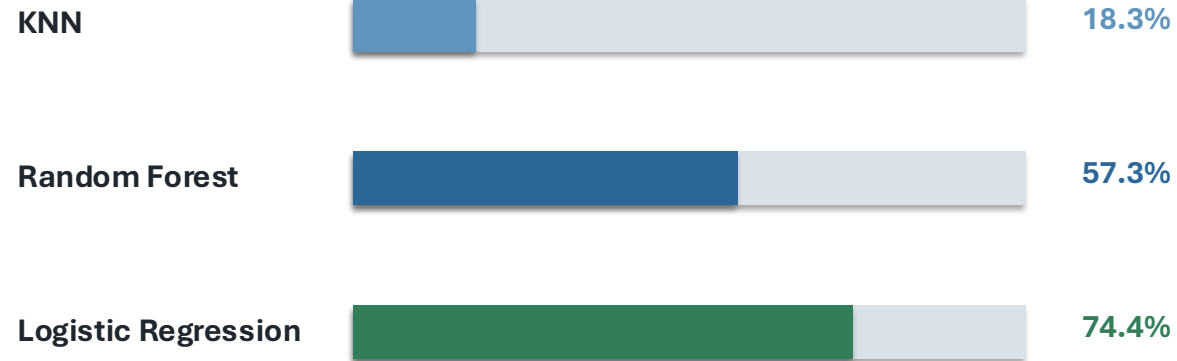


WINNER: LOGISTIC REGRESSION — highest test recall: 74.4%

6. Why Logistic Regression wins

Not the best on every metric, but the best fit for our decision

TEST RECALL



Important trade-off

Random Forest has better precision (48.0%) and F1 (52.2%).

But it finds fewer bankrupt companies: 57.3% recall vs 74.4%.

Logistic Regression
recall

Train 77.0% → CV 71.2% → Test
74.4%

Decision

Because recall was chosen before training, Logistic Regression is our recommended model.

Random Forest also shows stronger overfitting: train recall 95.7% vs CV 51.8%.

7. What does the winner get wrong?

Confusion matrix + consequences + safeguards

		Predicted	
		Operational	Bankrupt
Actual	Operational	875	213
	Bankrupt	21	61

21 FALSE NEGATIVES

At-risk companies are missed → possible missed support.

213 FALSE POSITIVES

Healthy companies are flagged → unnecessary concern/outreach.

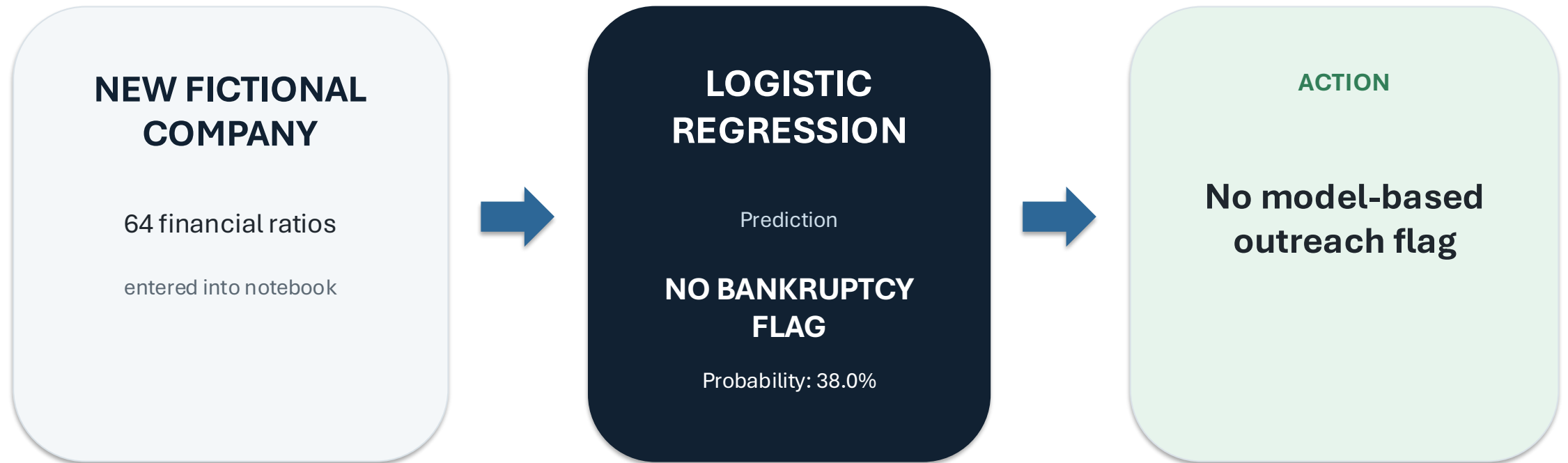
SAFEGUARDS

Confidential · voluntary support · human review · no automatic financial decisions

Limitation: demographic fairness cannot be established because the dataset lacks region, ownership demographics and complete industry information.

8. Live prediction

Working prototype: new input → model → output



CONCLUSION: Logistic Regression wins because 74.4% recall best matches our goal of missing as few at-risk companies as possible.

During the presentation: run the final fictional-company cell live in Colab.

Sources & Q&A

SOURCES

Statistics Poland (2023)
Registrations and bankruptcies of enterprises in Q1 2023

UCI Machine Learning Repository
Polish Companies Bankruptcy Data · CC BY 4.0

United Nations
SDG 8 — Decent Work and Economic Growth

scikit-learn
Model evaluation, pipelines and classifiers

PIPELINE DECISION TO DEFEND

Why recall?

- Bankruptcy is the minority class (~7%).
- Accuracy could look high while missing bankrupt companies.
- For Anna, a false negative means a company that may need support is missed.
- We therefore selected recall before training.

If asked “Why not Random Forest?” → better precision/F1, but lower recall (57.3%).